GCSE Mathematics - Foundation Tier

Paper 2 (Calculator) Practice - Set 2

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

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Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use pi = 3.142.

1.

Write these values in order of size. Start with the smallest.

```
1/3          0.3          30%          0.33
```

(Total for Question 1 is 1 marks)

2.

Work out 35% of £320.

(Total for Question 2 is 2 marks)

3.

(a) Convert 3.5 hours into minutes. (1)

(b) Convert 270 minutes into hours and minutes. (1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 45 : 60 in its simplest form.

(Total for Question 4 is 2 marks)

5.

A pictogram shows the number of cars sold at a small garage over five days. Each car symbol represents 4 cars.

```
Mon    [car][car][car]                (3 symbols)
Tue    [car][car][half-car]           (2.5 symbols)
Wed    [car][car][car][car][car]      (5 symbols)
Thu    [car][car]                     (2 symbols)
Fri    [car][car][car][half-car]      (3.5 symbols)
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(a) How many cars were sold on Wednesday? (1)

(b) Work out the total number of cars sold over the five days. (2)

(Total for Question 5 is 3 marks)

6.

Simplify $7x - 4 + 3x + 9$.

(Total for Question 6 is 2 marks)

7.

A solid wooden block has a volume of 250 cm^3 and is made of oak. The density of oak is 0.72 g/cm^3.

Work out the mass of the block. Give your answer in grams.

(Total for Question 7 is 3 marks)

8.

Here are the shoe sizes of 9 students.

4 5 5 6 6 6 7 8 9

(a) Write down the median shoe size. (1)

(b) Work out the range of the shoe sizes. (1)

(c) Write down the mode. (1)

(Total for Question 8 is 3 marks)

9.

A box of 84 sweets is shared between two children, Ella and Finn, in the ratio 3 : 4.

Work out the number of sweets that each child receives.

(Total for Question 9 is 3 marks)

10.

Solve $4(x + 3) = 28$.

You must show all your working.

(Total for Question 10 is 3 marks)

11.

The diagram shows an L-shape made from two rectangles. The L-shape has the following measurements: an outer rectangle 10 cm by 8 cm, with a rectangle of 6 cm by 5 cm removed from the top-right corner.

[Diagram: L-shape: outer rectangle 10 cm by 8 cm with a 6 cm by 5 cm rectangle removed from the top-right corner.]

Work out the total area of the L-shape. Give your answer in cm^2.

(Total for Question 11 is 3 marks)

12.

The pie chart shows how 120 people travel to work.

[Diagram: pie chart with sectors - Car: 180 deg, Bus: 90 deg, Train: 60 deg, Walk: 30 deg.]

(a) How many people travel to work by bus? (2)

(b) What percentage of people walk to work? (2)

(Total for Question 12 is 4 marks)

13.

The area A of a circle with radius r is given by A = pi x r^2.

(a) Work out the area when r = 6 cm. Give your answer to 1 decimal place. Use pi = 3.142. (2)

The cost C, in pounds, of hiring a van is given by C = 25 + 0.4m, where m is the number of miles driven.

(b) Work out the cost when m = 120 miles. (2)

(Total for Question 13 is 4 marks)

14.

A coat originally costs £85. It is reduced by 18% in a sale.

(a) Work out the sale price of the coat. (3)

Jay invests £2000 in an account paying 3% simple interest per year.

(b) Work out the total amount of interest Jay receives after 4 years. (2)

(Total for Question 14 is 5 marks)

15.

Here are the first four terms of an arithmetic sequence.

```
5,    8,    11,    14,    ...
```

(a) Write down the next term of the sequence. (1)

(b) Find an expression for the nth term of the sequence. (2)

(c) Is 100 a term in this sequence? You must show your working. (2)

(Total for Question 15 is 5 marks)

16.

Sara and Tom share some money in the ratio 2 : 5. Tom receives £40 more than Sara.

Work out the total amount of money they share.

(Total for Question 16 is 4 marks)

17.

A teacher records the height (cm) and arm span (cm) of 9 students.

Height	150	154	158	160	163	166	170	173	178
Arm span	148	152	157	160	162	165	171	172	179

(a) On the grid provided, plot these data as a scatter graph. (2)

(b) Describe the correlation shown by the data. (1)

(c) Draw a line of best fit. (1)

(d) A different student has a height of 168 cm. Use your line of best fit to estimate the arm span. (1)

(Total for Question 17 is 5 marks)

18.

A solid cylinder has a radius of 5 cm and a height of 12 cm.

[Diagram: solid cylinder of radius 5 cm and height 12 cm.]

(a) Work out the volume of the cylinder. Give your answer to 1 decimal place. Use pi = 3.142. (3)

The cylinder is made of brass with density 8.4 g/cm^3.

(b) Work out the mass of the cylinder. Give your answer in kilograms to 2 decimal places. (2)

(Total for Question 18 is 5 marks)

19.

The table shows the number of books read by each of 30 students in a month.

Books	0	1	2	3	4	5
Freq.	4	7	9	6	3	1

(a) Draw an accurate bar chart to represent this information. Label both axes. (3)

(b) Work out the mean number of books read per student. Give your answer to 2 decimal places. (2)

(Total for Question 19 is 5 marks)

20.

The sum of three consecutive integers is 105.

Let the smallest integer be x.

(a) Show that 3x + 3 = 105. (2)

(b) Solve the equation to find the value of x. (2)

(c) Write down the three integers. (1)

(Total for Question 20 is 5 marks)

21.

A supermarket sells two sizes of orange juice.

Small carton:	500 ml	for	£1.20
Large carton:	1.5 L	for	£3.30

(a) Work out the cost per 100 ml of the small carton. (2)

(b) Work out the cost per 100 ml of the large carton. (2)

(c) Which carton is the better value for money? Give a reason for your answer. (2)

(Total for Question 21 is 6 marks)

22.

A class of 60 students chose to study either French or Spanish. The two-way table shows some of the data.

	Boys	Girls	Total
French	14	___	30
Spanish	___	17	___
Total	___	___	60

(a) Complete the two-way table. (3)

(b) A student is chosen at random from the class. Work out the probability that the student is a girl who chose Spanish. (2)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 2 (Calculator) - Set 2 | 80 marks

Question 1

- Correct order: 0.3, 30%, 1/3, 0.33 - B1 (because 0.3 = 30%, 1/3 = 0.333..., 0.33 lies between)

Total: 1 marks

Question 2

- 320 x 0.35 or 320 x 35 / 100 - M1
- £112 - A1

Total: 2 marks

Question 3

- (a) 3.5 x 60 = 210 minutes - B1
- (b) 270 / 60 = 4 remainder 30 -> 4 hours 30 minutes - B1

Total: 2 marks

Question 4

- Both parts divided by a common factor (e.g. 15 : 20) - M1
- 3 : 4 - A1

Total: 2 marks

Question 5

- (a) 5 x 4 = 20 - B1
- (b) Total symbols 3 + 2.5 + 5 + 2 + 3.5 = 16; multiplied by 4 - M1
- (b) 64 (cars) - A1

Total: 3 marks

Question 6

- 10x or 5 collected correctly - M1
- 10x + 5 - A1

Total: 2 marks

Question 7

- Mass = density x volume - M1
- 0.72 x 250 - M1
- 180 (g) - A1

Total: 3 marks

Question 8

- (a) Median = 6 - B1
- (b) Range = 9 - 4 = 5 - B1
- (c) Mode = 6 - B1

Total: 3 marks

Question 9

- 84 / (3 + 4) = 12 - M1
- Either 3 x 12 or 4 x 12 worked out - M1
- Ella 36 and Finn 48 - A1

Total: 3 marks

Question 10

- Either x + 3 = 7 (dividing by 4) or 4x + 12 = 28 (expand) - M1
- Linear step to isolate x (e.g. 4x = 16) - M1
- x = 4 - A1

Total: 3 marks

Question 11

- Whole rectangle 10 x 8 = 80 cm^2 or split L into two rectangles - M1
- Removed rectangle 6 x 5 = 30 cm^2 or second part area - M1
- 50 (cm^2) - A1

Total: 3 marks

Question 12

- (a) 90 / 360 x 120 - M1
- (a) 30 (people) - A1
- (b) 30 / 360 x 100 - M1
- (b) 8.33% (accept 8.3% to 1dp) - A1

Total: 4 marks

Question 13

- (a) 3.142 x 6^2 (= 3.142 x 36) - M1
- (a) 113.1 (cm^2) accept 113.0 to 113.2 - A1
- (b) 25 + 0.4 x 120 - M1
- (b) £73 - A1

Total: 4 marks

Question 14

- (a) 18% of 85 = 15.30 or multiplier 0.82 - M1
- (a) 85 - 15.30 or 85 x 0.82 - M1
- (a) £69.70 - A1
- (b) 2000 x 0.03 x 4 or 60 x 4 - M1
- (b) £240 - A1

Total: 5 marks

Question 15

- (a) 17 - B1
- (b) Common difference 3 noted (3n + ...) - M1
- (b) 3n + 2 - A1
- (c) 3n + 2 = 100 -> n = 98/3 not an integer - M1
- (c) States 100 is NOT in the sequence with reason - A1

Total: 5 marks

Question 16

- Difference of parts = 5 - 2 = 3 parts - M1
- One part = 40 / 3 - M1
- Total parts = 7, so 7 x (40/3) - M1
- Total = £93.33 (accept £93.34 or exact fraction 280/3) - A1

Total: 4 marks

Question 17

- (a) All 9 points plotted correctly - B2 (B1 if 6 or more correct)
- (b) Positive (correlation, strong) - B1
- (c) Sensible line of best fit drawn - B1
- (d) Estimate from line at height 168 cm (accept 166 - 172) - B1

Total: 5 marks

Question 18

- (a) pi x 5^2 x 12 written or attempted - M1
- (a) 3.142 x 25 x 12 evaluated - M1
- (a) 942.6 (cm^3) accept 942.5 to 942.7 - A1
- (b) 942.6 x 8.4 (= 7917.84 g) and / 1000 - M1
- (b) 7.92 (kg) - A1

Total: 5 marks

Question 19

- (a) Axes correctly labelled (Books read / Frequency) with appropriate scale - B1
- (a) Heights drawn for 6 bars - B2 (B1 for 4 or 5 correct)
- (b) Sum(fx) = 0 + 7 + 18 + 18 + 12 + 5 = 60 - M1
- (b) Mean = 60 / 30 = 2.00 - A1

Total: 5 marks

Question 20

- (a) Three integers identified as x, x+1, x+2 - M1
- (a) x + (x+1) + (x+2) = 3x + 3 = 105 shown clearly - C1
- (b) 3x = 102 - M1
- (b) x = 34 - A1
- (c) 34, 35, 36 - B1

Total: 5 marks

Question 21

- (a) 1.20 / 5 (cost per 100 ml in small) - M1
- (a) £0.24 per 100 ml - A1
- (b) 1.5 L = 1500 ml and 3.30 / 15 (cost per 100 ml in large) - M1
- (b) £0.22 per 100 ml - A1
- (c) Comparison stated and large carton chosen - M1
- (c) Large carton is better value, with reason (lower cost per 100 ml) - A1

Total: 6 marks

Question 22

- (a) French girls = 30 - 14 = 16 - B1
- (a) Spanish total = 60 - 30 = 30; Spanish boys = 30 - 17 = 13 - M1
- (a) All totals consistent: boys total 27, girls total 33 - A1
- (b) 17 / 60 seen - M1
- (b) 17/60 - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 2 (Calculator) - Set 2 | Detailed explanations

Question 1

Express each value as a decimal so they can be compared.

1/3 = 0.333...

0.3 = 0.300

30% = 30/100 = 0.300

0.33 = 0.330

So 0.3 = 30% (equal), then 0.33, then 1/3.

Order: 0.3, 30%, 0.33, 1/3. (0.3 and 30% are equal; you can list them either way provided both come before 0.33.)

Question 2

35% as a decimal = 0.35.

0.35 x 320 = 112.

Answer: £112.

Mental check: 10% of 320 = £32, so 30% = £96, and 5% = £16. Total £112.

Question 3

(a) 1 hour = 60 minutes, so 3.5 hours = 3.5 x 60 = 210 minutes.

(b) 270 / 60 = 4.5, i.e. 4 whole hours and 0.5 of an hour.

0.5 hour = 30 minutes, so 4 hours 30 minutes.

Question 4

HCF of 45 and 60. Factors of 45: 1, 3, 5, 9, 15, 45. Factors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. HCF = 15.

Divide both by 15: 45/15 = 3, 60/15 = 4.

Answer: 3 : 4.

Question 5

(a) 5 symbols on Wednesday and each symbol = 4 cars, so 5 x 4 = 20 cars.

(b) Add the symbols across the week: 3 + 2.5 + 5 + 2 + 3.5 = 16 symbols.

Total cars = 16 x 4 = 64.

Question 6

Collect like terms.

x-terms: 7x + 3x = 10x.

Constants: -4 + 9 = +5.

Answer: $10x + 5$.

Question 7

Mass = density x volume.

Mass = 0.72 x 250 = 180.

Answer: 180 g.

Question 8

(a) Median is the middle value of the 9 sorted values. The 5th value is 6.

(b) Range = largest - smallest = 9 - 4 = 5.

(c) Mode is the most common value. 6 appears three times, the most.

Question 9

Total parts = 3 + 4 = 7. One part = 84 / 7 = 12 sweets.

Ella: 3 x 12 = 36 sweets.

Finn: 4 x 12 = 48 sweets.

Check: 36 + 48 = 84.

Question 10

Method 1 - divide by 4 first.

$4(x + 3) = 28 \rightarrow x + 3 = 7 \rightarrow x = 4$.

Method 2 - expand first.

$4x + 12 = 28 \rightarrow 4x = 16 \rightarrow x = 4$.

Check: $4(4 + 3) = 4 \times 7 = 28$.

Question 11

Split the L-shape into two rectangles or subtract the missing rectangle.

Method - subtract: outer rectangle 10 x 8 = 80 cm^2; missing rectangle 6 x 5 = 30 cm^2; L-shape = 80 - 30 = 50 cm^2.

Answer: 50 cm^2.

Question 12

(a) Bus sector = 90 deg out of 360 deg.

People = (90 / 360) x 120 = 0.25 x 120 = 30.

(b) Walk sector = 30 deg.

Percentage = (30 / 360) x 100 = 8.33...%, i.e. 8.3% to 1 dp.

Question 13

(a) $A = \pi \times r^2 = 3.142 \times 6^2 = 3.142 \times 36 = 113.112$.

Rounded to 1 dp: 113.1 cm^2.
(b) C = 25 + 0.4 x 120 = 25 + 48 = 73.
Answer: £73.

Question 14

(a) 18% of £85 = 0.18 x 85 = £15.30.
Sale price = 85 - 15.30 = £69.70. (Or 0.82 x 85 = £69.70.)
(b) Simple interest = principal x rate x time = 2000 x 0.03 x 4 = £240.

Question 15

(a) Common difference is +3. Next term = 14 + 3 = 17.
(b) The nth term = (common difference) x n + (zeroth term).
Zeroth term = 5 - 3 = 2. So nth term = 3n + 2.
(c) If 100 were in the sequence then 3n + 2 = 100, giving n = 98/3 = 32.66...
n is not a whole number, so 100 is NOT a term of the sequence.

Question 16

The difference between Tom and Sara is 5 parts - 2 parts = 3 parts.
3 parts = £40, so 1 part = 40 / 3 = £13.333... .
Total = 7 parts = 7 x 40/3 = £280/3 = £93.33 (to the nearest penny).

Question 17

(a) Plot 9 points (height on x-axis, arm span on y-axis).
(b) Points rise from lower-left to upper-right -> positive correlation (and strong).
(c) Draw a straight line through the middle of the points.
(d) Read off the y-value at x = 168 cm. Typical answer 166 - 172 cm.

Question 18

(a) Volume of cylinder = pi x r^2 x h = 3.142 x 5^2 x 12 = 3.142 x 25 x 12 = 942.6 cm^3.
(b) Mass = density x volume = 8.4 x 942.6 = 7917.84 g.
Convert to kg: 7917.84 / 1000 = 7.91784 kg, i.e. 7.92 kg to 2 dp.

Question 19

(a) Draw axes (Number of books on x-axis, Frequency on y-axis) and 6 bars of heights 4, 7, 9, 6, 3, 1.
(b) Mean from frequency table: Sum(fx) = 0x4 + 1x7 + 2x9 + 3x6 + 4x3 + 5x1 = 0 + 7 + 18 + 18 + 12 + 5 = 60.
Total students Sum(f) = 30. Mean = 60 / 30 = 2.00.

Question 20

(a) Three consecutive integers: x, $x+1$, $x+2$.

Sum = $x + (x+1) + (x+2) = 3x + 3$.

Given sum is 105, so $3x + 3 = 105$. (Shown.)

(b) Subtract 3: $3x = 102$. Divide by 3: $x = 34$.

(c) Integers are 34, 35, 36. Check: $34 + 35 + 36 = 105$.

Question 21

(a) Small carton: 500 ml is 5 lots of 100 ml. Cost per 100 ml = 1.20 / 5 = £0.24.

(b) Large carton: 1.5 L = 1500 ml is 15 lots of 100 ml. Cost per 100 ml = 3.30 / 15 = £0.22.

(c) £0.22 < £0.24, so the large carton is cheaper per 100 ml.

Conclusion: the large (1.5 L) carton is the better value.

Question 22

(a) French total is 30. French boys = 14, so French girls = 30 - 14 = 16.

Spanish total = 60 - 30 = 30. Spanish girls = 17, so Spanish boys = 30 - 17 = 13.

Boys total = 14 + 13 = 27. Girls total = 16 + 17 = 33. Check: 27 + 33 = 60.

(b) P(girl AND Spanish) = (Spanish girls) / total = 17 / 60.
